

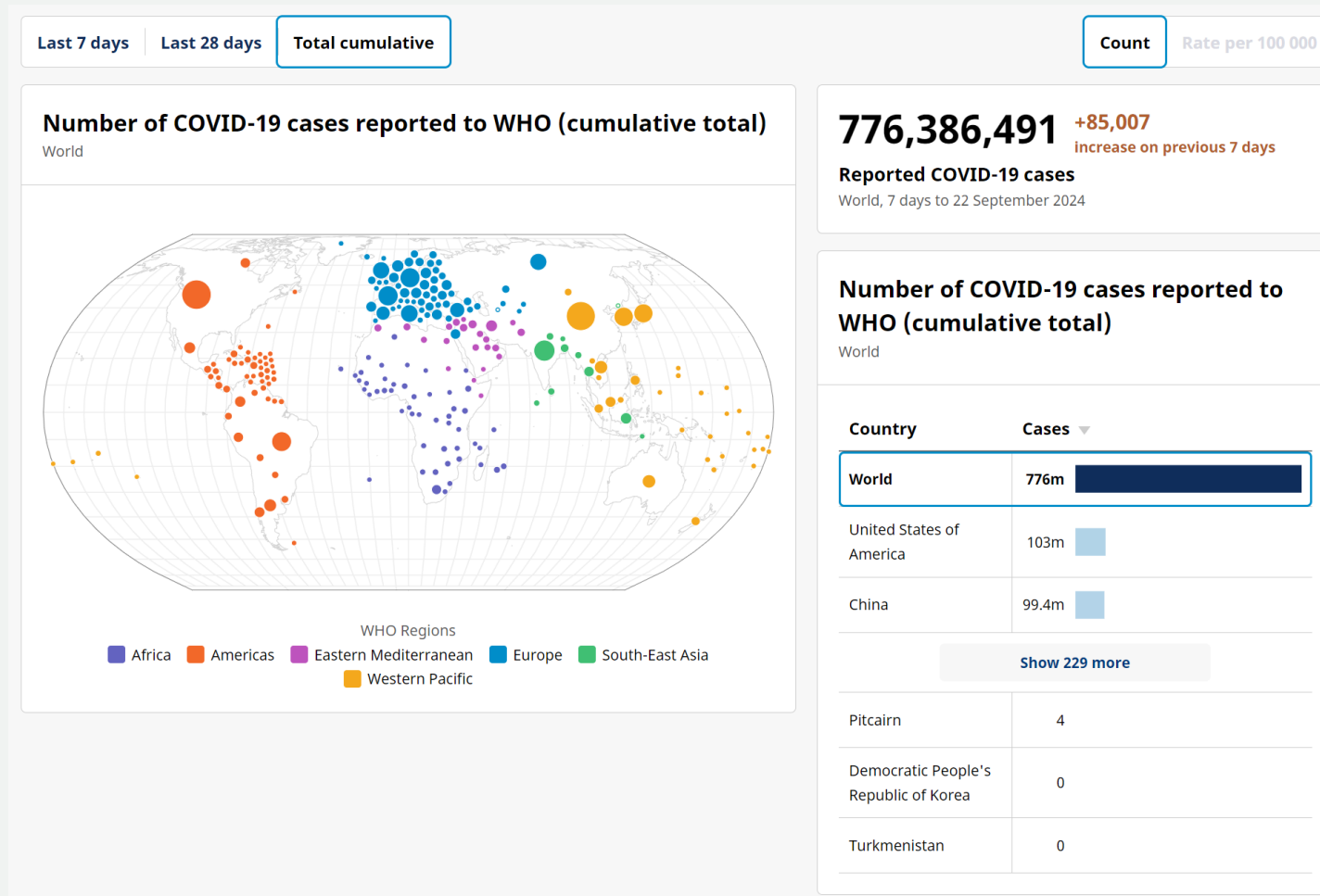
VISUALISING THE IMPACT OF COVID-19

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MXB362 Advanced Data Visualisation and Data
Science
Case Study Presentation



CURRENT PANDEMIC VISUALISATIONS



Most Visualisations of Covid-19

- Provide total case numbers
- Regional Breakdowns

Good for providing a quantitative overview of the pandemic.

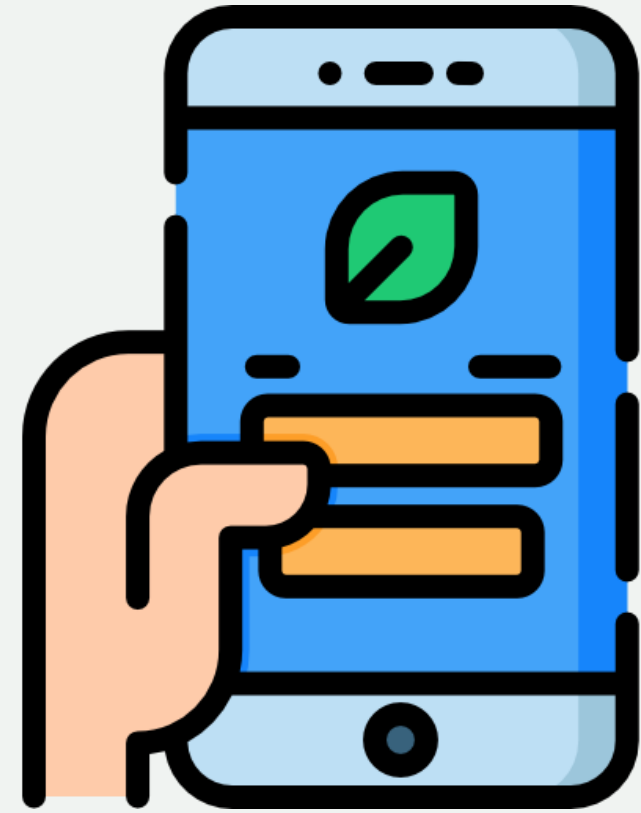
However, they often fail to communicate the urgency or real-time severity of the situation to the general public.

Fig 1. WHO Covid-19 Dashboard (World Health Organization, n.d.)

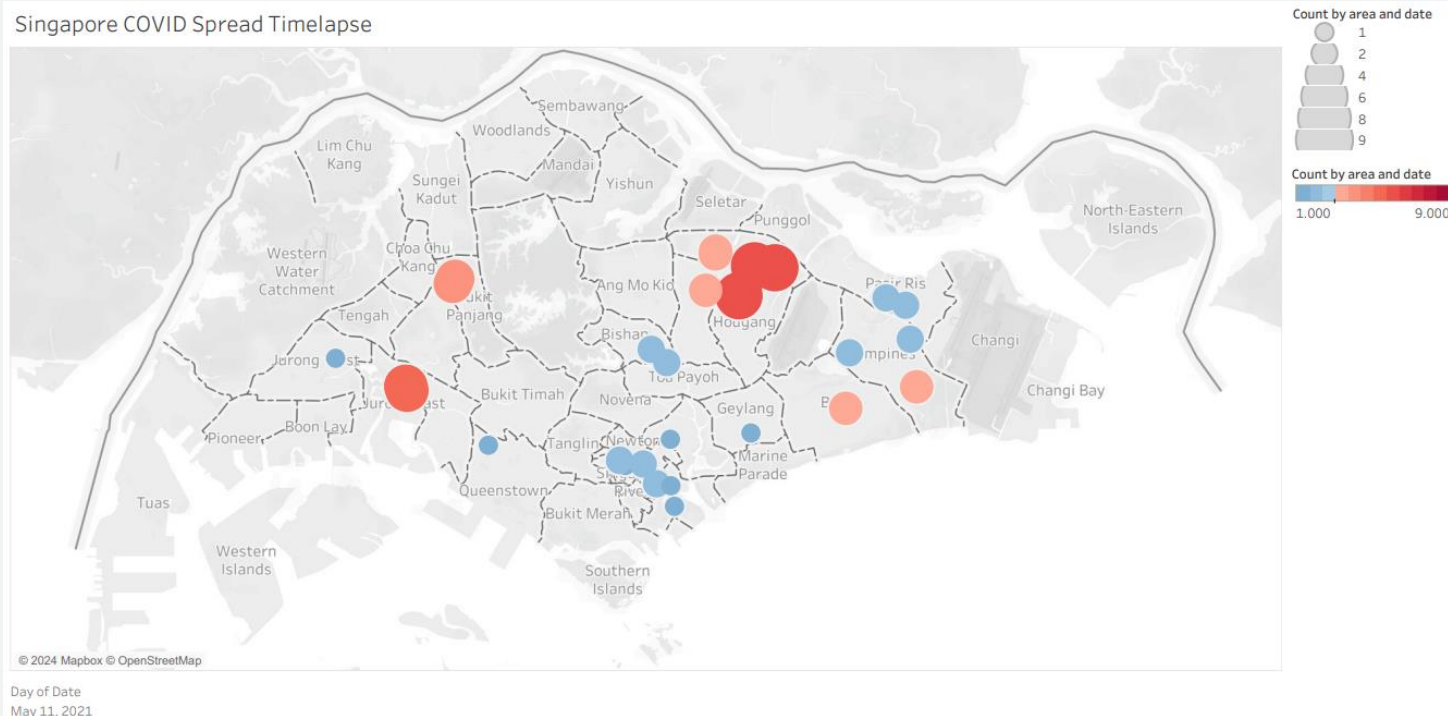
CASE STUDY MOTIVATIONS

Develops a visualisation tool that delivers up-to-date, daily insights on the local spread of the pandemic

1. The tool should be user-friendly, making it easy for the public to understand current risks and trends in their area.
2. Empower them to make informed decisions.



VISUALISING SG COVID-19 SPREAD (SINGAPORE EXAMPLE)



This visualisation is an animation that shows the daily distribution of COVID-19 cases in Singapore

- Unable to tell how increase of cases affects increase in size of circle
- Circle is so big that it covers the entire district, mislead audience to think the entire estate is a hotspot
- Unnecessary usage of 2 colours

Fig 2. Singapore COVID Spread Book (Chua, 2021)

DATA SOURCE

The dataset used is a list that includes public places COVID-19 cases had visited for more than 30 minutes (Chua, 2020).

The main data that will be visualised is the following:

- Time
- Date
- Latitude and Longitude Coordinates
- Area (Estate names)



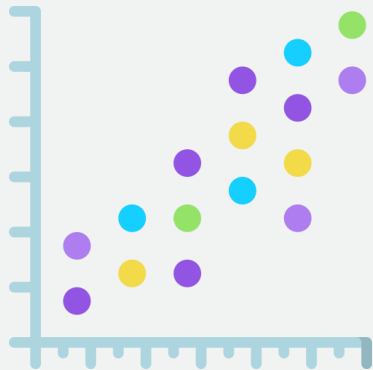
Dataset used: SG Covid Locations (SG-COVID-LOCATIONS, 2021)

VISUALISATION TECHNIQUES AND TOOLS

Technique 1: Choropleth Map



Technique 2: Scatter Plot



Tools used

1. Data Manipulation: R
2. Visualisation: Python
3. Development on: Jupyter Notebook
4. Video Editing: Microsoft Clipchamp

BACKGROUND RESEARCH

- Self-monitoring for up to 5 days is advised for close contacts of potential COVID-19 cases (Gov.SG | Updates to Health Protocols, n.d.).
- Following government protocols, the produced visualisation should be an animation displaying COVID-19 case locations over a 5-day period.



RESULTS AND OUTPUTS

Data Manipulations

1. Extracting Relevant Information from GEOJson 'Description' field
2. Merged location data with geographic boundaries data based on area names.
3. Aggregated data by date and name, counting occurrences

```
# Loading Singapore's Map in GeoJson
geojson_file = "MasterPlan2019PlanningAreaBoundaryNoSea.geojson"
gdf = gpd.read_file(geojson_file)
gdf.crs = "EPSG:4326"
print(gdf.Description[0])
```

```
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RESULTS AND OUTPUTS

Visualisation Technique 1: Animated Choropleth Map (Matplotlib)

Used Matplotlib to produce visualisation instead of other libraries that offer more interactivity.

Interactive elements like zooming, panning, and hovering over data points are not necessary when target audience is the general public whose primary need is to understand the spread and density.

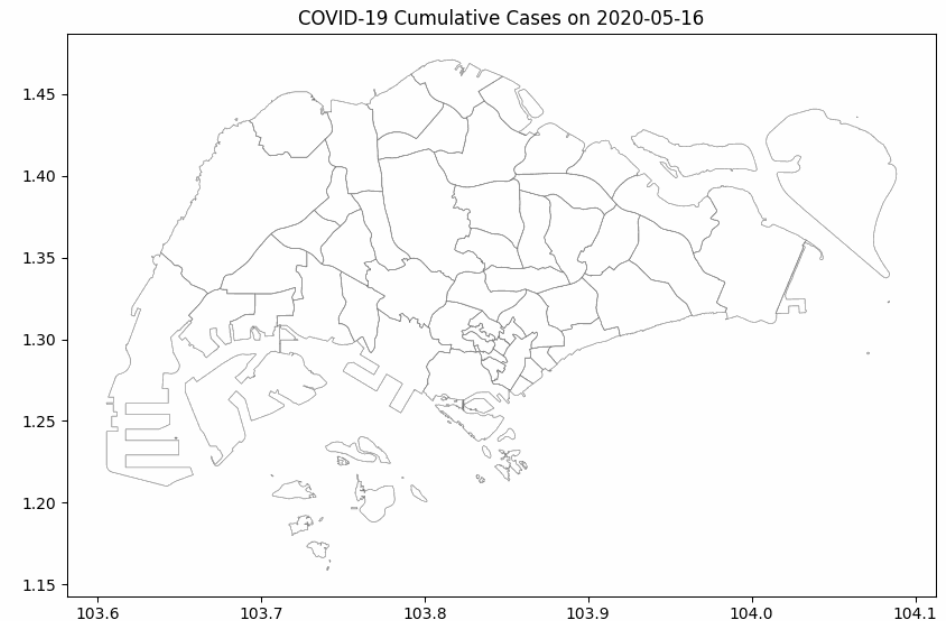


Fig 3. SG Covid-19 Choropleth Map Animation

RESULTS AND OUTPUTS

Visualisation Technique 2: Animated Scatter Plot (Matplotlib)

Experimented with different ways to animate with the locations of COVID-19 cases across Singapore

1. Sequential Animation of Individual Cases
2. Rolling 5-Day Window Animation of Individual Cases
3. Rolling 5-Day Window Animation of Daily Cases

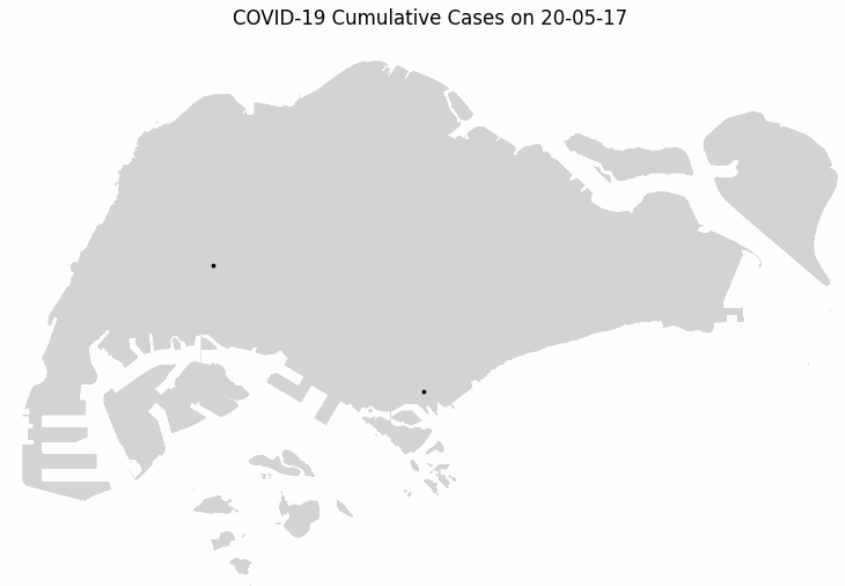


Fig 4. SG Covid-19 Scatter Plot Animation (Method 3)

RESULTS AND OUTPUTS

Combination of Both Technique

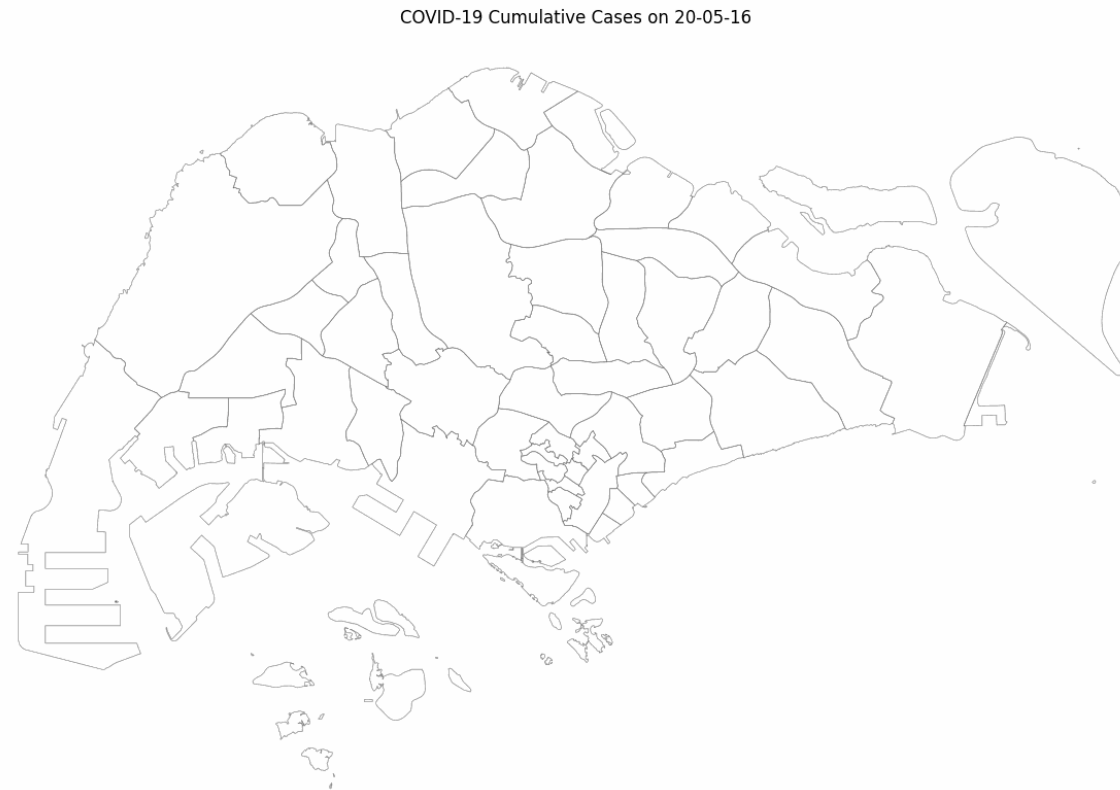


Fig 5 . SG Covid-19 Combination of Techniques

EXEGESIS OF THE EFFECTIVENESS OF VISUALISATION OUTPUTS

1. Lines

Used thin lines to separate different areas of Singapore

2. Colours

Used of monochromatic colour scheme

3. Clear Labels

Clear and concise labels, limited to the legend and a title indicating the cumulative COVID-19 cases and the corresponding date.

4. Negative Space

Use of negative space to ensure that the visualisation stands out

5. Minimised Complexity

Minimises complexity by focusing solely on essential elements: the geographic areas, the cumulative case counts, and the date.

PROBLEMS



- Initial animation frames show misalignment between the scatter plot and choropleth map. Potential cause: mismatch between the initial dataset and the GIS map of Singapore.
- Simplified focus on Singapore's geographic areas, case counts, and date makes the visualisation suitable for local audiences.
Limits effectiveness for international viewers who may not recognize specific areas or districts.



FUTURE WORK

- Incorporate live feeds from reliable sources (health organisations and government databases).
- Integrate real-time data into the visualisation tool to enhance relevance and usability during pandemics.



REFERENCES

- World Health Organization. (n.d.). COVID-19 cases | WHO COVID-19 dashboard. data.who. <https://data.who.int/dashboards/covid19/cases>
- Chua, H. X. (2021, June 30). Data VIS Singapore Jun 2021 MeetUp: Visualising SG COVID-19 Spread [Video]. YouTube. <https://www.youtube.com/watch?v=OFRegA91bAM>
- Chua, H. S. (n.d.). GitHub - huishun98/SG-COVID-data-automated: This project prepares data for a visualisation showing the spread of COVID-19 in Singapore across time. GitHub. <https://github.com/huishun98/SG-COVID-data-automated?tab=readme-ov-file>
- Chua, H. S. (2021, June 5). Singapore COVID Spread Book. Tableau Public. <https://public.tableau.com/app/profile/hui.shun/viz/SingaporeCOVIDSpreadBookAutoUpdate/Dashboard>
- Chua, H. X. (2020, May 30). Public Places Visited by Singapore Covid-19 Cases. data.world. <https://data.world/hxchua/public-places-visited-in-singapore-by-covid-19-cases>
- SG-COVID-Locations. (n.d.). Google Docs. <https://docs.google.com/spreadsheets/d/19EPRvGyAMnYZn9LwfFa4UqYrt-MQ94-qEUg76vbl2Gs/edit?gid=0#gid=0>
- Gov.SG | Updates to health protocols. (n.d.). Default. <https://www.gov.sg/article/updates-to-health-protocols>
- MXB362. (n.d.). Google Docs. https://docs.google.com/spreadsheets/d/1ZSmeVb8H-TFp9U5e9KdF5GGieY2Bom53Z6UJeNp_jmw/edit?usp=sharing
- Urban Redevelopment Authority [URA]. (2024, June). Master Plan 2019 Planning Area Boundary (No Sea). data.gov.sg. https://data.gov.sg/datasets/d_4765db0e87b9c86336792efe8a1f7a66/view